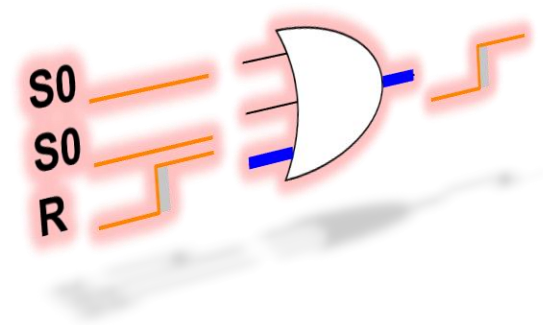


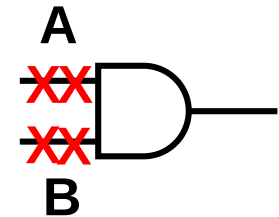
Delay Test

- Introduction and delay fault models
- Path Delay Fault
- **Transition Delay Fault**
 - ◆ **Fault Simulation**
 - ◆ **Test Generation**
- Experimental Results* (not in exam)
- Issues of Delay Tests* (not in exam)
- Conclusions



TDF Detection

- How to detect A STR fault ?
 - ♦ V_1 : Control A to 0
 - ♦ V_2 : Detect A *stuck-at 0*
- How to detect A STF fault ?
 - ♦ V_1 : Control A to 1
 - ♦ V_2 : Detect A *stuck-at 1*
- TDF assumes delay fault size very large
 - ♦ **No requirement** about propagation path

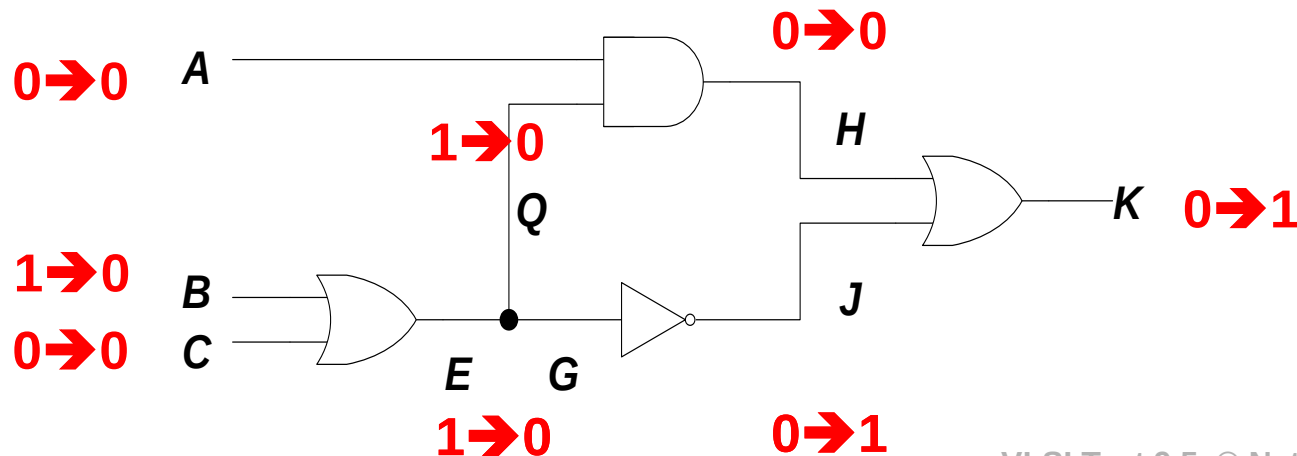


V_1 AB	V_2 AB	Detected input TDF
0X	11	A STR
1X	01	A STF
X0	11	B STR
X1	10	B STF

TDF Very Similar to SSF

TDF Fault Simulation

- Transition delay fault simulation can be built on SSF simulation
 - ♦ Just **add additional check for V_1**
 - ♦ V_2 is same as SSF simulation
- Example: $V_1 ABC = 010$, $V_2 ABC = 000$
 - ♦ V_2 detects B SA1, C SA1, E SA1, G SA1, J SA0, K SA0
 - ♦ $\{V_1, V_2\}$ Detected transition faults
 - * **B STF, E STF, G , STF, J STR, K STR**
 - * C STF not detected because no transition
 - * Q STF not detected because Q SA1 not detected by V_2



Quiz

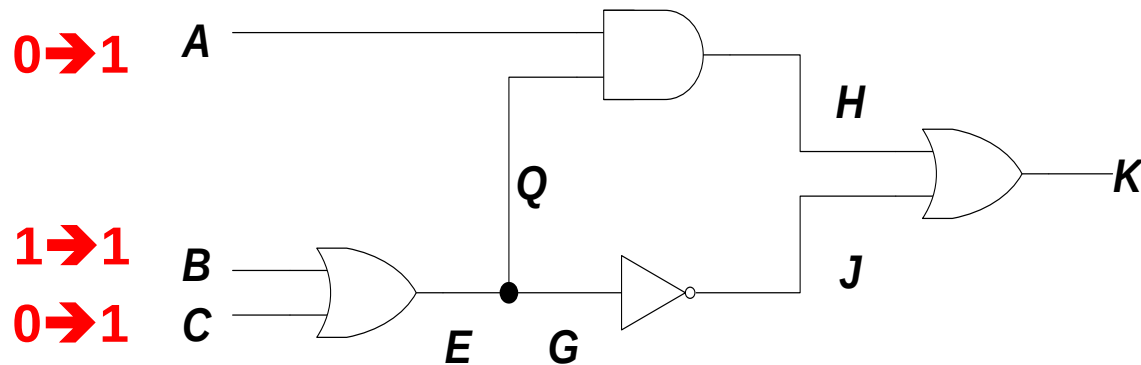
Q: Given this 2-patter test, which transition faults are detected?

V₁ ABC= 010

V_2 ABC= 111

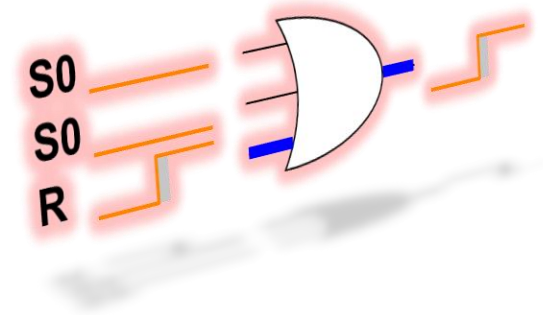
considers *A* STR, *H* STR, *C* STR faults

A:



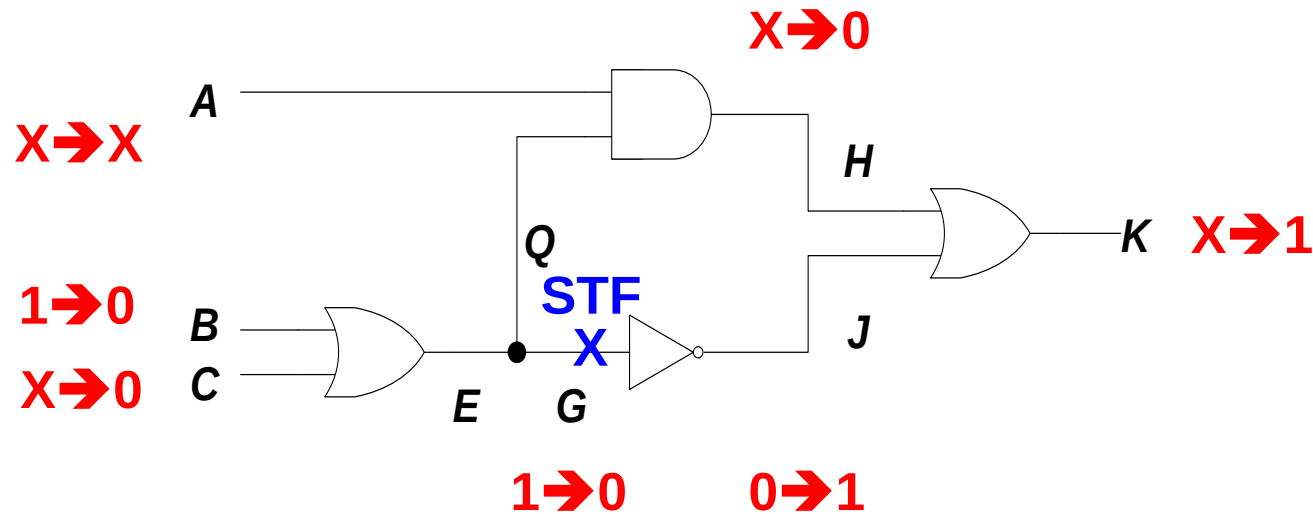
Delay Test

- Introduction and delay fault models
- Path Delay Fault
- **Transition Delay Fault**
 - ◆ **Fault Simulation**
 - ◆ **Test Generation**
- Delay Test Application
- Circuit Model for Delay Test ATPG
- Experimental Results* (not in exam)
- Issues of Delay Tests* (not in exam)
- Conclusions



TDF ATPG

- First SSF for V_2 , then control V_1
- Example: use PODEM for *G slow-to-fall (STF) fault*
 - ♦ V_2 objective: detect G stuck-at 1 $\rightarrow B_2=0$
 - ♦ V_2 objective: detect G stuck-at 1 $\rightarrow C_2=0$, V_2 generated
 - ♦ V_1 objective: $G = 1 \rightarrow B_1=1$
 - ♦ Test generated: $V_1 ABC = X1X$, $V_2 ABC = X00$



Generate V_2 , then V_1

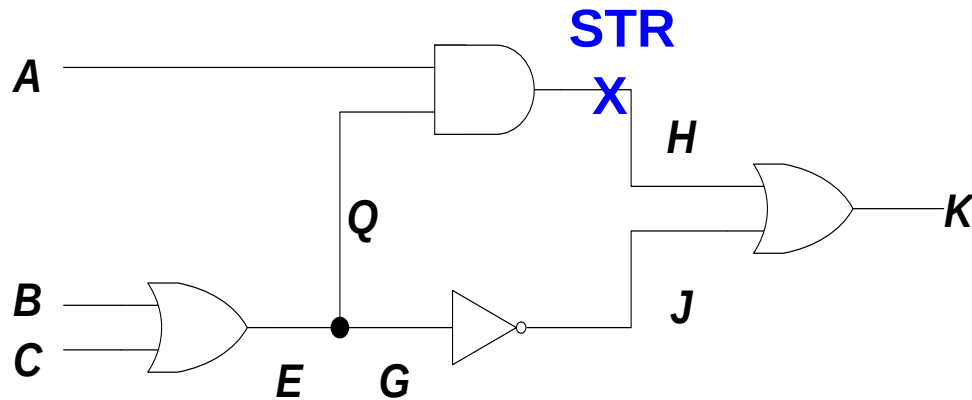
Quiz

Q: Generate a test for H STR fault

A:

V_1 ABC=

V_2 ABC=

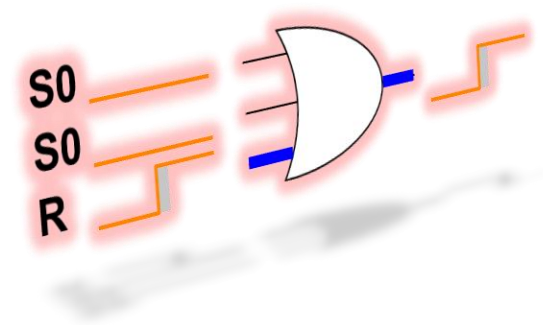


Pros and Cons of TDF

- **Advantages:**
 - + Leverage on mature SSF ATPG algorithm
 - + High coverage
 - + ATPG easy because no attention to path
- **Disadvantages**
 - SSF ATPG tends to trace short paths
 - ◇ may not effective for small delay defect (SDD)
- **TDF test patterns currently “most basic” delay testing**

Summary

- Transition Delay Fault
 - ◆ Fault Simulation
 - * Based on SSF fault simulation, **check V_1**
 - ◆ Test Generation
 - * Generate SSF for **V_2**
 - * Then control opposite value at **V_1**



FFT

- Q: why generate V_2 first, then V_1 . (why not V_1 first, then V_2)
- Example: use PODEM for G slow-to-fall (STF) fault
 - ♦ V_2 objective: detect G stuck-at 1 $\rightarrow B_2=0$
 - ♦ V_2 objective: detect G stuck-at 1 $\rightarrow C_2=0$, V_2 generated
 - ♦ V_1 objective: $G = 1 \rightarrow B_1=1$
 - ♦ Test generated: $V_1 ABC = X1X$, $V_2 ABC = X00$

